Grade: Check Plus

✓ Qa

Assignment 1

Stats, Dec 27
Ben Newman

x	$f(x)$	$xf(x)$	$(x-\mu)^2 f(x)$
10	.10	1	10
20	.15	3	60
40	.3	12	480
60	.15	9	540
80	.25	20	1600
100	.05	5	500

a. $P[x \geq 60]$:

60 80 100
.15 .25 .05 = .45 = $\boxed{45\%}$

✓ Qb

b. $1+3+12+9+20+5 = \boxed{\$50}$

$\sigma^2 = 3190 - (50)^2 = \sqrt{690} = \boxed{26.27}$

✓ Qc

c. Expected value ≠ actual ~~demonstration~~ $\sqrt{\sigma^2 = 3190}$
it is the statistical average, which $\sigma = 51.24$
does not correspond to withdraw amounts with
the ATM.

d. $xf(x) = 50 - 26.27 \leq \quad \leq 50 + 26.27$

$\sigma = 26.27 \quad 23.73 \leq$ ✓ Qd ≤ 76.27

40 60
.3 + .15 = .45 = $\boxed{45\%}$

e. $50 - 2(26.27) \leq \quad \leq 50 + 2(26.27)$

$50 - 52.54 = -2.54 \leq$ ✓ Qe $\leq 102.54 \quad 52.54$

$10 - 100 = 1 = \boxed{100\%}$